The Derivative of the Sigmoid Function

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Where this fits: Pipeline map step 8 (Model). See the Course map.

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- **Builds on:** Sigmoid function.
- **Compare with:** Tanh.

1. Overview

Key point: The slope of the S-curve is steepest in the middle and nearly flat at both ends. Its exact value comes from the curve's own height: slope = height × (1 − height).

Gradient descent (G-862) on the **log loss** (G-303; the measure of how wrong the predicted probabilities are) needs the **derivative** (G-595) of the **sigmoid function** (G-1798). The derivative is the slope of the curve: how fast its output changes when its input z changes a little. The same derivative appears again in neural networks, where a neuron can use the sigmoid to produce its output. This short Note derives it once.

In plain words, the answer is easy to picture. The sigmoid is an S-shaped curve: flat on the far left, steep in the middle, flat again on the far right. So its slope is close to 0 at both ends and largest in the middle (Figure 3 in section 4 rides a **tangent line** (G-1945) along the curve). The formula of this Note gives that slope exactly.

As a reminder, the sigmoid is

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The sigmoid squeezes any number into the range 0 to 1 (see the sigmoid function). Here e is Euler's number, $e \approx 2.718$, and e^{-z} is 1 divided by e^{z}.

2. Two rules we need

Key point: The chain rule, and the **derivative** (G-595) of e^(−z), which is −e^(−z).

Notation. $\frac{d}{dz}$ means "the slope with respect to z": how much the quantity after it changes when z grows by a tiny amount. Take $u = 1 + e^{-z}$, a number that depends on z. At $z = 0$:

$$u = 1 + e^{0} = 1 + 1 = 2$$

At $z = 1$:

$$u = 1 + e^{-1} = 1 + 0.368 = 1.368$$

The symbol $\frac{du}{dz}$ is the slope of u itself.

Chain rule (G-371): to differentiate a function of a function, differentiate the outer one, keep the inside, and multiply by the derivative of the inside. Here the outer function is the reciprocal $1/u$ and the inner function is u. Worked at $z = 0$:

Step	Value
u at $z = 0$	2
$\frac{du}{dz}$ at $z = 0$ (the slope of $1 + e^{-z}$ is $-e^{-z}$, by the exponential rule below)	$-e^0 = -1$
Outer slope: $-\frac{1}{u^2}$	$-\frac{1}{4} = -0.25$
Multiply outer by inner	$-0.25 \times (-1) = 0.25$

The result 0.25 is the slope of $1/u$ with respect to z at $z = 0$. In general form, for any u:

$$\frac{d}{dz}\left(\frac{1}{u}\right) = -\frac{1}{u^2} \cdot \frac{du}{dz}$$

Check: take $u = 2$ and $du/dz = -1$. The formula gives:

$$-\frac{1}{4} \times (-1) = 0.25$$

This is the same number.

Exponential: the slope of e^z is e^z itself. By the chain rule, the slope of e^{-z} is e^{-z} times the slope of the inside $-z$, which is -1. At $z = 1$:

$$e^{-1} = 0.368$$

$$\text{slope} = 0.368 \times (-1) = -0.368$$

In general form:

$$\frac{d}{dz}e^{-z} = -e^{-z}$$

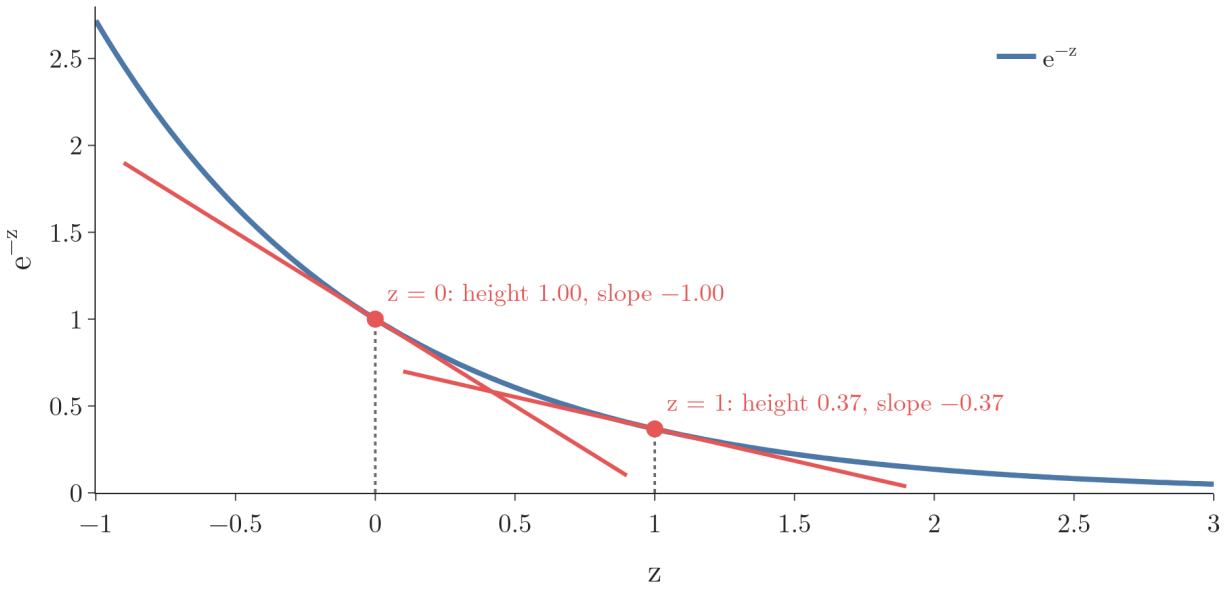


Figure 1: The curve e^{-z} with its tangent lines at $z = 0$ and $z = 1$. Each slope is the curve's own height with a minus sign.

Each red line in Figure 1 is a **tangent line** (G-1945): the straight line that touches the curve at one point and has the curve's slope there. Compare each red label's height and slope: they match except for the sign, which is the rule $\frac{d}{dz}e^{-z} = -e^{-z}$.

3. The derivation

> **Key point:** Differentiate, then split the fraction into σ(z) times $(1 - \sigma(z))$.

3.1 Differentiate

> **Key point:** Treat σ(z) as 1/u with u = 1 + e^(−z).

Write $u = 1 + e^{-z}$, so $\sigma(z) = 1/u$. The **derivative** (G-595) of u is:

$$0 + (-e^{-z}) = -e^{-z}$$

Using the reciprocal rule:

$$\sigma'(z) = -\frac{1}{(1+e^{-z})^2} \cdot \left(-e^{-z}\right)$$

$$= \frac{e^{-z}}{(1+e^{-z})^2}$$

The two minus signs cancel. Check at $z = 0$ ($u = 2$, $e^0 = 1$):

$$\sigma'(0) = \frac{e^0}{(1+e^0)^2}$$

$$= \frac{1}{2^2} = 0.25$$

3.2 Split into two fractions

> **Key point:** One fraction is σ(z) itself; the other turns out to be $1 - \sigma$(z).

Split the result into a product of two fractions:

$$\sigma'(z) = \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}}$$

At $z = 0$ the two fractions are each 0.5:

$$\frac{1}{1+1} = 0.5$$
$$0.5 \times 0.5 = 0.25$$

This is the same slope. The first fraction is $\sigma(z)$. For the second, add and subtract 1 in the numerator:

$$\frac{e^{-z}}{1+e^{-z}} = \frac{(1+e^{-z}) - 1}{1+e^{-z}}$$

$$= 1 - \frac{1}{1+e^{-z}}$$

$$= 1 - \sigma(z)$$

3.3 The result

> **Key point:** σ'(z) = σ(z)(1 − σ(z)).

Putting the two pieces together:

$$\boxed{\sigma'(z) = \sigma(z)\left(1 - \sigma(z)\right)}$$

With numbers: at $z = 2$, $\sigma(2) = 0.881$, so:

$$\sigma'(2) = 0.881 \times 0.119 = 0.105$$

Figure 3 (Section 4) draws the two factors as the blue and orange parts of a bar of height 1.

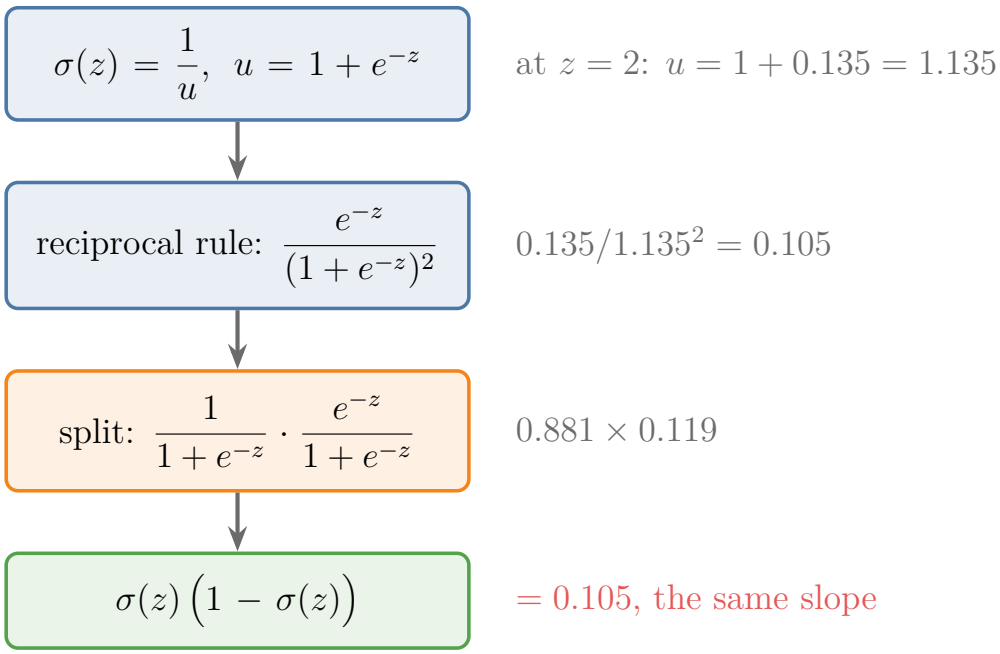


Figure 2: The derivation in four steps, each checked with numbers at $z = 2$: the raw derivative and the product $\sigma(1 - \sigma)$ both give 0.105.

In Figure 2, follow the numbers beside the boxes: the messy fraction and the neat product land on the same slope, 0.105.

The form $\sigma(1 - \sigma)$ is very convenient. When a model has already computed $\sigma(z)$ for its prediction, the **derivative** (G-595) costs just one subtraction and one multiplication.

4. The shape of the derivative

> **Key point:** The **derivative** (G-595) is largest, 0.25, at z = 0 and falls towards 0 on both sides.

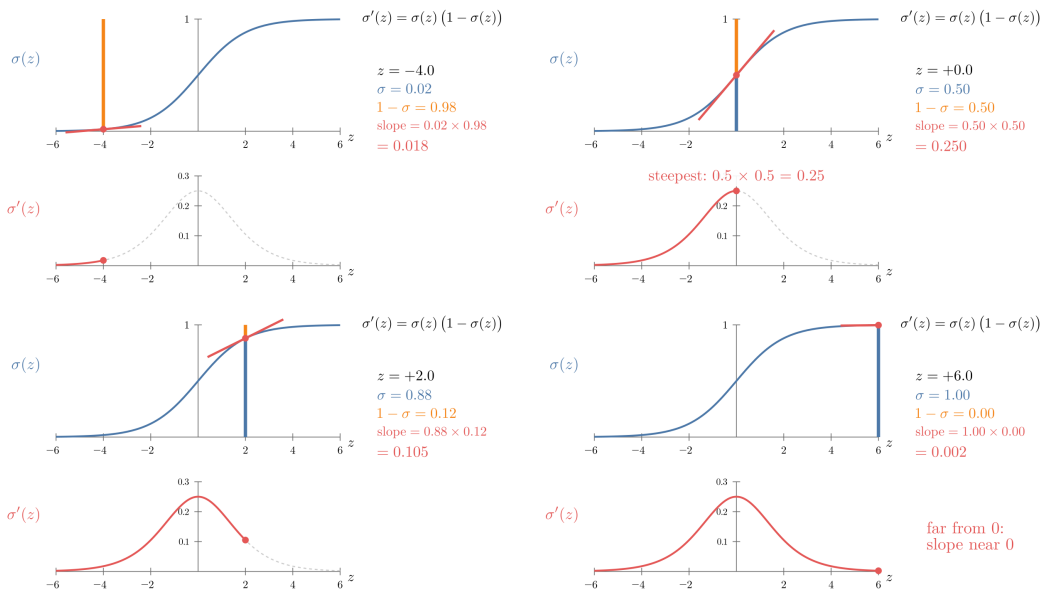


Figure 3: A tangent line (red) rides the sigmoid; its slope is traced below. The bar at z splits into $\sigma(z)$ (blue) and $1-\sigma(z)$ (orange), and the slope is their product. Key frames: $z = -4$, 0, 2, 6

In Figure 3, watch the two coloured parts of the bar: the slope is large only when both parts are large, which happens at $z = 0$, where each is 0.5. Reading the derivative as the slope of a **tangent line** (G-1945) that moves along the curve follows Sanderson's *Essence of Calculus* (3Blue1Brown, "The paradox of the derivative").

z	-4	-2	0	2	4
$\sigma(z)$	0.018	0.119	0.5	0.881	0.982
$\sigma'(z)$	0.018	0.105	0.25	0.105	0.018

- At $z = 0$ the sigmoid is steepest. The value 0.25 is the largest value the derivative can take:

$$0.5 \times 0.5 = 0.25$$

- Far from 0, the sigmoid is almost flat, near 0 or 1, so its slope is almost 0.
- The curve is symmetric: $\sigma'(-z) = \sigma'(z)$.

What the shape means. The slope says how much the output $\sigma(z)$ moves when z is nudged. Near $z = 0$ the model is unsure (output near 0.5), and a small change in z changes the output a lot. Far from 0 the output is already close to 0 or 1, and the same nudge changes it by almost nothing: at $z = 4$ the slope is 0.018, about 14 times smaller than at $z = 0$.

The derivative is used in two places later:

- gradient descent for logistic regression puts it into the **chain rule** (G-371) to get the gradient of the log loss;

- the vanishing gradient problem shows what the small slopes do in a deep neural network (the Extra below gives the short version).

Python: The derivative, checked numerically.

```
def sigmoid(z):
    return 1 / (1 + np.exp(-z))

def sigmoid_derivative(z):
    return sigmoid(z) * (1 - sigmoid(z))

h = 1e-5
(sigmoid(2 + h) - sigmoid(2 - h)) / (2 * h)    # 0.104994
sigmoid_derivative(2)                           # 0.104994
```

The numerical slope (rise over a tiny run) matches the formula to six decimals.

Extra: Because the derivative is at most 0.25, and almost 0 for large or small z, stacking many sigmoid layers (steps that each pass their output to the next) in a deep neural network multiplies many small numbers together. The gradients become tiny and learning stalls: the **vanishing gradient** (G-2070) problem. Saturating sigmoids (flat at both ends) make gradient-based learning very difficult, so the sigmoid is now discouraged inside the hidden layers of a network (the layers between input and output), and **ReLU (a function that outputs 0 for negative inputs and the input itself otherwise)** (G-1668) is the default choice (Goodfellow et al. §6.1, §6.3.2).

5. Summary

- The sigmoid:
$$\sigma(z) = 1/(1 + e^{-z})$$
- Its derivative, found with the **chain rule** (G-371) and one algebra step:
$$\sigma'(z) = \sigma(z)(1 - \sigma(z))$$
This form is cheap, because a model that has already computed $\sigma(z)$ gets the slope with one subtraction and one multiplication.
- The **derivative** (G-595) peaks at 0.25 when $z = 0$ and approaches 0 far from it, because the S-curve is steepest in the middle and flat at both ends; so a nudge to z moves an unsure output a lot and a confident output hardly at all.
- The next Note uses it to derive the gradient of the log loss, which **gradient descent** (G-862) needs to train logistic regression.

6. Sources

Built from

- CampusX, "Derivative of Sigmoid Function", YouTube, https://www.youtube.com/watch?v=awjXaFR
- Sanderson, G. (3Blue1Brown), "The paradox of the derivative", *Essence of Calculus*, chapter 2. 3blue1brown.com/lessons/derivatives

Other references

- **Goodfellow et al.:** Goodfellow, I., Bengio, Y. and Courville, A. *Deep Learning*. MIT Press, 2016. Sections 6.1 and 6.3.2.

7. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Chain rule (G-371)	To differentiate a function of a function, multiply the outer derivative by the inner derivative.
Gradient descent (G-862)	Finding the lowest point of a function by repeated small steps downhill.
Binary cross entropy (log loss) (G-303)	The loss function of logistic regression for two classes: per row $-y \log a - (1 - y) \log(1 - a)$, averaged over the rows (the average cross entropy).

Term	Meaning
Derivative (G-595)	The slope of a function at a point: how fast the output changes as the input changes there. It is zero at the lowest point of a smooth curve, which is how training finds a minimum.
Sigmoid function (G-1798)	An S-shaped function that squashes any number into the range 0 to 1, $\sigma(z) = 1/(1 + e^{-z})$; it turns a score into a probability, as in logistic regression.
Tangent line (G-1945)	The line that touches a curve at one point with the curve's slope there; the limit of secant lines.
Vanishing gradient (G-2070)	Gradients shrinking towards 0 as they pass through many layers, which slows learning.
ReLU (G-1668)	Rectified linear unit, the activation $\max(0, z)$: it passes a positive input unchanged and turns a negative one into 0; the usual default activation for hidden layers.